

MASTER'S COMPREHENSIVE EXAMINATION

Part II

Applied (90 minutes)

December 4, 2010

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. The Body mass Index BMI of a person is $\text{Weight} / (\text{Height})^2$ where weight is in Kg and height is in meters. BMI measures Weight adjusted for Height.

A. Suppose for country X, all of the following are true for Height and BMI of Adults

- * Height is uniformly distributed on 1.50 to 2.10 Meters
- * Height is independent of BMI
- * The population mean for BMI is 23.0

Then what is the mean Weight of an adult in country X?

B. Now someone believes Adult Height and BMI are not independent in country X and has the following data for means of BMI by different ranges of Height

Height Category (Range in M)	Mean BMI (in Kg)
[1.50 – 1.65) n =20 adults	24
[1.65 – 1.80) n= 20 adults	22
[1.80 – 1.95) n=20 adults	26
[1.95 – 2.10) n=20 adults	20

Assume that the variance of BMI is the same for all height categories and the pooled estimate of this variance (with $80 - 4 = 76$ df) is 56.5

Use the best test (which most efficiently adjusts for multiple comparisons) of

$$H_0: \mu_{[1.50-1.65)} = \mu_{[1.65-1.80)} = \mu_{[1.80-1.95)} = \mu_{[1.95-2.10)}$$

Against

H_a : For at least one of the categories, the mean BMI is different

2. A random sample of 80 men and 120 women infected with HIV is obtained, some (but not all) of whom have AIDS. All subjects have HIV viral load (HVL) measured in a blood sample. The following analyses are done.

I. Comparison of HVL between Men and Women

Gender	Mean of HVL	Std-Dev of HVL
Men (n=80)	5.05	2.07
Women (n=120)	4.13	1.05

II. Logistic Regression Models of having AIDs (Yes, No)

II-1. Unadjusted Association of Female Gender with having AIDS

Variable	Beta	Std-Err
Female Vs. Male	0.02	0.20

II-2. Multivariate Association of Female Gender and HVL with having AIDS

Variable	Beta	Std-Err
Female Vs. Male	-0.53	0.22
HVL (per Unit)	0.55	0.24

→
OVER

Use $P=0.05$ for all questions do not adjust for multiple comparisons.

- a. Is there statistical evidence that the mean of HVL differs between men and women in the population that produced this sample? State any assumptions you need to make to answer this?
- b. Give an approximate 95 percent confidence for the ratios of the odds for a man to a woman in the population with the same viral load to have AIDS.
- c. Is there statistical evidence that the overall probability of men and women from the population that generated this sample to have AIDs is different

3. An experiment is designed where 3 machines are randomly selected and each machine is assigned to a different group of 3 operators selected at random. Each operator uses the machine 2 times at different periods during a given week. The amounts of time needed to produce 1 item (completion time) given in standard units are shown below:

Machine	1			2			3		
Operator	1	2	3	4	5	6	7	8	9
	13.2	14.1	13.8	9.5	12.6	9.7	17.4	16.0	15.4
	14.3	14.6	12.7	9.8	11.7	11.2	17.6	13.0	14.4

- (a) Calculate the appropriate quantities (SS, df, MS, F) and write down the standard ANOVA table.
 - (b) Test at $\alpha=0.05$ if there are differences in completion times among the machines (show details).
 - (c) Test at $\alpha=0.05$ if there are differences in completion times between operators within the machines (show details).
4. (a) A gambler has a fair coin (with one head and one tail) and a two-headed coin. He selects one of the coins at random, and when he flips it, it shows heads. What is the probability that the coin he selected is the fair coin?
- (b) Suppose that he flips the same coin a second time and again it shows heads. Then he flips the coin 2 more times. What is the expected number of heads across all 4 flips and the variance of the number of heads across all 4 flips?